

NUCLEAR POWER AND THE THREE-STAGE PROGRAMME

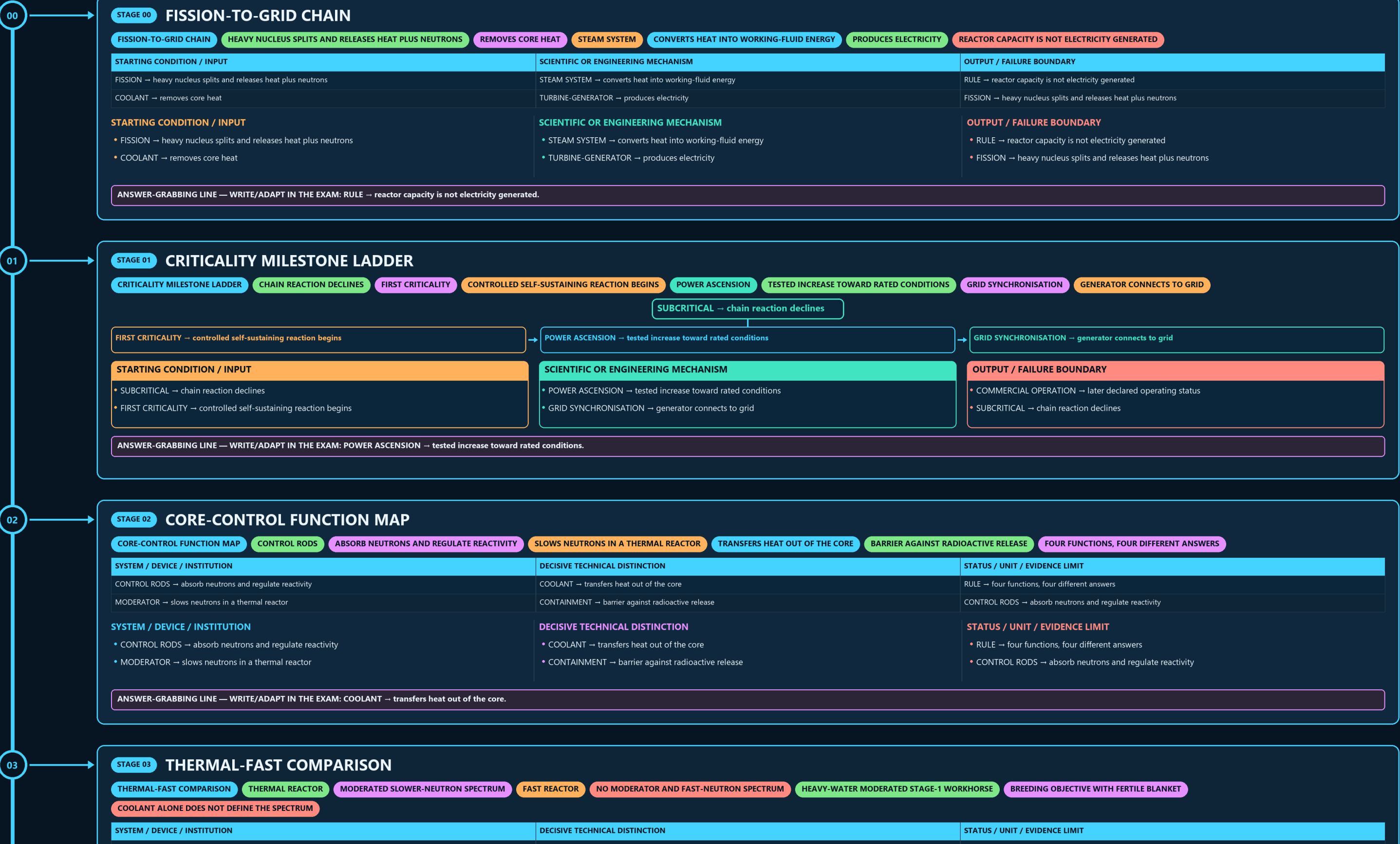
FISSION-TO-GRID CHAIN → CRITICALITY MILESTONE LADDER → CORE-CONTROL FUNCTION MAP → THERMAL-FAST COMPARISON → FISSILE-FERTILE CONVERSION MAP → THREE-STAGE PROGRAMME RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



SYSTEM / DEVICE / INSTITUTION

- CONTROL RODS → absorb neutrons and regulate reactivity
- MODERATOR → slows neutrons in a thermal reactor

DECISIVE TECHNICAL DISTINCTION

- COOLANT → transfers heat out of the core
- CONTAINMENT → barrier against radioactive release

STATUS / UNIT / EVIDENCE LIMIT

- RULE → four functions, four different answers
- CONTROL RODS → absorb neutrons and regulate reactivity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOLANT → transfers heat out of the core.

03

STAGE 03THERMAL-FAST COMPARISON

THERMAL-FAST COMPARISONTHERMAL REACTORMODERATED SLOWER-NEUTRON SPECTRUMFAST REACTORNO MODERATOR AND FAST-NEUTRON SPECTRUMHEAVY-WATER MODERATED STAGE-1 WORKHORSEBREEDING OBJECTIVE WITH FERTILE BLANKET

COOLANT ALONE DOES NOT DEFINE THE SPECTRUM

| SYSTEM / DEVICE / INSTITUTION                         | DECISIVE TECHNICAL DISTINCTION                 | STATUS / UNIT / EVIDENCE LIMIT                      |
|-------------------------------------------------------|------------------------------------------------|-----------------------------------------------------|
| THERMAL REACTOR → moderated slower-neutron spectrum   | PHWR → heavy-water moderated Stage-1 workhorse | RULE → coolant alone does not define the spectrum   |
| FAST REACTOR → no moderator and fast-neutron spectrum | FBR → breeding objective with fertile blanket  | THERMAL REACTOR → moderated slower-neutron spectrum |

SYSTEM / DEVICE / INSTITUTION

- THERMAL REACTOR → moderated slower-neutron spectrum
- FAST REACTOR → no moderator and fast-neutron spectrum

DECISIVE TECHNICAL DISTINCTION

- PHWR → heavy-water moderated Stage-1 workhorse
- FBR → breeding objective with fertile blanket

STATUS / UNIT / EVIDENCE LIMIT

- RULE → coolant alone does not define the spectrum
- THERMAL REACTOR → moderated slower-neutron spectrum

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → coolant alone does not define the spectrum.

04

STAGE 04FISSILE-FERTILE CONVERSION MAP

FISSILE-FERTILE CONVERSION MAPFISSILE MATERIALSNEUTRON CAPTUREPU-239 PATHWAYNEUTRON CAPTURE AND DECAYU-233 PATHWAYFERTILE MATERIALNEEDS IRRADIATION AND CONVERSION

U-235 / PU-239 / U-233 → fissile materials

U-238 → neutron capture → PU-239 pathway

TH-232 → neutron capture and decay → U-233 pathway

FERTILE MATERIAL → needs irradiation and conversion

STARTING CONDITION / INPUT

- U-235 / PU-239 / U-233 → fissile materials
- U-238 → neutron capture → PU-239 pathway

SCIENTIFIC OR ENGINEERING MECHANISM

- TH-232 → neutron capture and decay → U-233 pathway
- FERTILE MATERIAL → needs irradiation and conversion

OUTPUT / FAILURE BOUNDARY

- RULE → resource abundance is not ready reactor fuel
- U-235 / PU-239 / U-233 → fissile materials

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → resource abundance is not ready reactor fuel.

05

STAGE 05THREE-STAGE PROGRAMME RAIL

THREE-STAGE PROGRAMME RAILSTAGE 1PHWR, NATURAL URANIUM AND PLUTONIUM-BEARING SPENT FUELSEPARATES MATERIAL FOR THE BREEDER BRIDGESTAGE 2FAST BREEDER AND FISSILE-INVENTORY MULTIPLICATIONTHORIUM IRRADIATIONPREPARES U-233 ROUTE

STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

REPROCESSING → separates material for the breeder bridge

STAGE 2 → fast breeder and fissile-inventory multiplication

THORIUM IRRADIATION → prepares U-233 route

STAGE 3 → long-term thorium-U-233 utilisation horizon

CONCEPT / ARCHITECTURE

- STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel
- REPROCESSING → separates material for the breeder bridge

MECHANISM / APPLICATION

- STAGE 2 → fast breeder and fissile-inventory multiplication
- THORIUM IRRADIATION → prepares U-233 route

READINESS / STATUS LIMIT

- STAGE 3 → long-term thorium-U-233 utilisation horizon
- STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STAGE 2 → fast breeder and fissile-inventory multiplication.

06

STAGE 06PFBR STATUS CARD

PFBR STATUS CARDIGCAR FAST-REACTOR AND SODIUM EXPERTISEBHAVINI PROJECT RESPONSIBILITY500 MWE DESIGN VALUE IN FETCHED DAE TEXT6 APRIL 2026FIRST CRITICALITY ACHIEVEDNOT ESTABLISHEDGRID SYNCHRONISATION OR COMMERCIAL OPERATION

DESIGN → IGCAR fast-reactor and sodium expertise

BUILD / COMMISSION → BHAVINI project responsibility

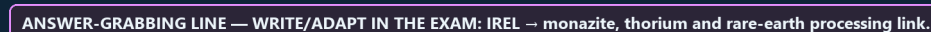
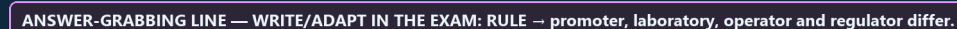
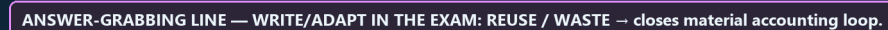
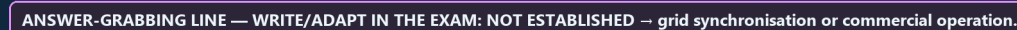
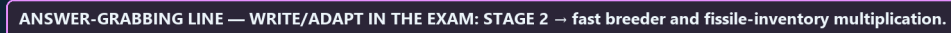
RATING → 500 MWe design value in fetched DAE text

6 APRIL 2026 → first criticality achieved

STARTING CONDITION / INPUT

SCIENTIFIC OR ENGINEERING MECHANISM

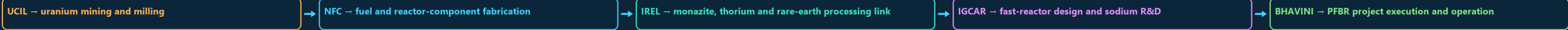
OUTPUT / FAILURE BOUNDARY



09

STAGE 09 FUEL-AND-PROJECT CHAIN

FUEL-AND-PROJECT CHAIN URANIUM MINING AND MILLING FUEL AND REACTOR-COMPONENT FABRICATION MONAZITE, THORIUM AND RARE-EARTH PROCESSING LINK FAST-REACTOR DESIGN AND SODIUM R&D PFBR PROJECT EXECUTION AND OPERATION



STARTING CONDITION / INPUT

- UCIL → uranium mining and milling
- NFC → fuel and reactor-component fabrication

SCIENTIFIC OR ENGINEERING MECHANISM

- IREL → monazite, thorium and rare-earth processing link
- IGCAR → fast-reactor design and sodium R&D

OUTPUT / FAILURE BOUNDARY

- BHAVINI → PFBR project execution and operation
- UCIL → uranium mining and milling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IREL → monazite, thorium and rare-earth processing link.

10

STAGE 10 REACTOR TAXONOMY

REACTOR TAXONOMY HEAVY WATER AND NATURAL-URANIUM STAGE-1 ROLE LIGHT-WATER SYSTEMS WITH DIFFERENT STEAM CYCLES FAST SPECTRUM AND BREEDING OBJECTIVE SMALL-REACTOR LINES DESIGN OR DEVELOPMENT STATUS TYPE AND DEPLOYMENT STATUS MUST BE STATED TOGETHER

| SYSTEM / DEVICE / INSTITUTION                               | DECISIVE TECHNICAL DISTINCTION                            | STATUS / UNIT / EVIDENCE LIMIT                            |
|-------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| PHWR → heavy water and natural-uranium Stage-1 role         | FBR → fast spectrum and breeding objective                | RULE → type and deployment status must be stated together |
| PWR / BWR → light-water systems with different steam cycles | AHWR / SMALL-REACTOR LINES → design or development status | PHWR → heavy water and natural-uranium Stage-1 role       |

SYSTEM / DEVICE / INSTITUTION

- PHWR → heavy water and natural-uranium Stage-1 role
- PWR / BWR → light-water systems with different steam cycles

DECISIVE TECHNICAL DISTINCTION

- FBR → fast spectrum and breeding objective
- AHWR / SMALL-REACTOR LINES → design or development status

STATUS / UNIT / EVIDENCE LIMIT

- RULE → type and deployment status must be stated together
- PHWR → heavy water and natural-uranium Stage-1 role

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → type and deployment status must be stated together.

11

STAGE 11 NUCLEAR ANSWER SPINE

NUCLEAR ANSWER SPINE FISSION, CRITICALITY, FISSILE, FERTILE AND FUEL CYCLE STAGE 1, PLUTONIUM BRIDGE, STAGE 2 AND THORIUM HORIZON DAE BARC NPCIL AERB IGCAR BHAVINI AND FUEL CHAIN CAPACITY CRITICALITY GRID GENERATION AND LEGAL STATUS

DATED FLEET FUEL REACTOR LAW COST AND SCHEDULE EVIDENCE



CONCEPT / ARCHITECTURE

- DEFINE → fission, criticality, fissile, fertile and fuel cycle
- TRACE → Stage 1, plutonium bridge, Stage 2 and thorium horizon

MECHANISM / APPLICATION

- MAP → DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain
- SEPARATE → capacity criticality grid generation and legal status

READINESS / STATUS LIMIT

- VERIFY → dated fleet fuel reactor law cost and schedule evidence
- DEFINE → fission, criticality, fissile, fertile and fuel cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated fleet fuel reactor law cost and schedule evidence.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT POLICY, STRATEGIC DIRECTION AND OVERALL NUCLEAR-GOVERNANCE UMBRELLA. R&D BACKBONE FOR REACTOR TECHNOLOGY, FUEL CYCLE AND ADVANCED THE FAST-REACTOR AND SODIUM-TECHNOLOGY DESIGN CENTRE — IT DESIGNED

BHAVINI IS THE SEPARATE COMPANY THAT BUILT IT AND CONFLATING IGCAR WITH BHAVINI IS A COMMON INSTITUTIONAL ERROR. COMMERCIAL REACTOR OPERATOR/DEVELOPER FOR THE CIVILIAN FLEET ALSO THE VEHICLE FOR THE BHARAT SMALL REACTOR (220

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- DAE: policy, strategic direction and overall nuclear-governance umbrella.
- BARC: R&D backbone for reactor technology, fuel cycle and advanced concepts (including the AHWR design and the BSMR line).
- IGCAR: the fast-reactor and sodium-technology design centre — it designed PFBR
- BHAVINI is the separate company that built it and will operate it.

READINESS, UNIT OR STATUS LIMIT

- Conflating IGCAR with BHAVINI is a common institutional error.
- NPCIL: commercial reactor operator/developer for the civilian fleet
- also the vehicle for the Bharat Small Reactor (220 MWe PHWR for captive industrial use) solicitation.
- BHAVINI: Stage 2 breeder-reactor implementation body linked to PFBR.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- NFC / UCIL / IREL: fuel fabrication, uranium mining, and monazite processing for
- thorium and rare earths — the physical supply chain beneath the three-stage narrative.
- AERB: technical-safety regulator, distinct from liability legislation — and, as of the verified position, not a statutorily independent authority .
- CAUTION: A major debate is how to balance ambitious capacity expansion with

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.